

NeoGravity Series: Independent Verification Suite

Companion computational materials for **The Theory of NeoGravity & Ether Dynamics** (J. S. Guagliardo, 2026).

Purpose

Every central mathematical claim in the series is reduced here to executable code. A reviewer can verify the entire derivation chain in under a minute rather than checking algebra by hand.

Running

```
pip install sympy numpy
python3 verify_neogravity.py
```

What is checked

#	Claim	Method
1	Hydrostatic equilibrium solution	SymPy <code>dsolve</code> on the first-order ODE
2	Exponential is not a 1/r power law	Series expansion at large r
3	Calibration $w = 1/4$	Solve first-order coefficient against measurement
4	Second-order divergence 2.00 vs 1.75	Taylor expansion of both refractive indices
5	Light deflection, three routes	Closed-form evaluation
6	Mercury perihelion	Standard secular-advance formula
7	Redshift and Shapiro delay	Closed-form evaluation
8	Frame dragging: topology vs	Explicit statement of what is fitted

	prefactor	
9	PTA correlation functions	Hellings-Downs and scalar-transverse evaluated
10	SMBHB orbital frequencies	Kepler's third law across parameter space
—	Dimensional consistency	Unit audit of every composite quantity

Epistemic status, stated plainly

The suite distinguishes three categories and prints the classification for each result:

- **DERIVED** — follows from the postulates with no free parameters: equilibrium profile, refractive index form, perihelion advance, gravitational redshift, Shapiro delay.
- **CALIBRATED** — fixed by matching one observation:
 - $w = 1/4$ from the measured solar light deflection
 - R_{eff} from the measured frame-dragging amplitude
- **TOPOLOGY ONLY** — shared functional form, physically distinct mechanism: the frame-dragging dipole $\sin(\theta)/r^2$.

The series has two fitted constants, not one. Any statement to the contrary is incorrect and is flagged by this suite.

Known corrections enforced here

1. $\exp(-A/r)$ is an exponential whose *exponent* depends on $1/r$. It is not a $1/r$ power law. The $1/r$ perturbation appears only at first order.
2. Stokes creeping flow (viscous, no-slip boundary) and the Kerr vacuum solution share dipole topology but not physical mechanism. The amplitude is matched, not derived.
3. Earth's moment of inertia is $I = 0.3307 M R^2$ (centrally condensed), not $(2/5) M R^2$. Canonical $R_{\text{eff}} = 4920 \text{ m}$.

Constants

CODATA 2018 and IAU nominal values, listed at the top of the script.

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